Review Report

Technical Paper Review Form

Paper ID: T007
Paper Title: Coordinated Optimization of Emergency Supply Distribution and Multi-Type Personnel Evacuation under Hazardous-Material Vehicle Incidents on Urban Expressways: A Case Study of Beijing

Comments to authors (within 300 words)

The topic presented in the paper is valid and important. In the reviewed paper, the Authors presented the problem of coordinated optimization of emergency supply distribution and multi-type personnel evacuation under hazardous-material vehicle incidents on urban expressways - as a case study of Beijing. Using a hypothetical chlorine release at Beijing's Sixth Ring Road as a scenario, this study develops an integrated emergency-response framework. The Authors used the Gaussian plume model and ERPG thresholds to delineate three-tier hazard zones and co-optimized a truck-drone logistics network and a multi-type evacuation fleet encompassing five modes and three vulnerability classes. Results show that incorporating shelter-in-place demand increases supply tasks from 40 to 70. Under lightly damaged road conditions, the truck-only scheme outperforms the truck-drone scheme, highlighting the relay delivery's applicability boundary. ERPG-based zoning reduces the active evacuation population from 235,000 to 13,800. Vulnerable groups are 100% evacuated, but critically injured victims achieve only a 2.8% evacuation rate, exposing a structural bottleneck in medical fleets.

The paper is well-structured and well-organized. The paper content is suitable for the conference scope.

RECOMMENDATIONS

Overall recommendation

☒ Accept

☐ Accept after minor revision (without re-review)

☐ To be revised and resubmitted (major revision)

☐ Reject

PRESENTATION RECOMMENDATION

☒ Oral presentation

☐ Poster presentation